

CO2 AT2 – Case Study

Name: R.Mouni

Reg. No.: 192472283

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Case Study 12: Customer Support Automation Using Generative AI and Large Language Models (LLMs)

Introduction

Customer support is one of the most important functions of any organization. It directly influences customer satisfaction, loyalty, and brand reputation. Traditional customer support systems rely heavily on human agents to answer customer queries through phone calls, emails, live chat, and social media. As businesses grow, handling a large volume of customer requests becomes challenging. Customers also expect quick, accurate, and 24/7 responses, which increases the workload on support teams.

Generative Artificial Intelligence (Generative AI) and Large Language Models (LLMs) such as GPT have transformed customer service by enabling intelligent virtual assistants and chatbots that can understand natural language, generate human-like responses, and resolve customer issues efficiently.

This case study presents a complete solution for implementing an AI-powered customer support automation system using modern Generative AI technologies.

Problem Statement

A multinational e-commerce company receives more than **60,000 customer support requests every day** through multiple channels such as:

- Website chat
- Mobile application
- Email
- WhatsApp
- Social media

The customer support team faces several challenges:

- Long waiting time during peak hours
- High operational costs
- Repetitive questions consume agent time

- Inconsistent responses from different agents
- Difficulty providing support 24×7
- Handling multilingual customer requests

The company wants an AI-powered customer support automation system that can:

- Answer customer queries instantly.
- Resolve common issues automatically.
- Escalate complex problems to human agents.
- Improve customer satisfaction.
- Reduce operational costs.

Objectives

The primary objectives of the proposed solution are:

- Automate customer interactions.
- Reduce average response time.
- Provide 24/7 customer support.
- Improve customer experience.
- Reduce workload on support agents.
- Increase resolution accuracy.
- Support multiple languages.
- Learn continuously from customer feedback.

Scenario Analysis

The customer support department handles different types of requests every day.

Examples include:

Customer Query	Expected Response
Where is my order?	Provide tracking information
How can I return my product?	Explain return policy
Payment failed	Suggest payment troubleshooting
Product damaged	Create replacement request
Refund status	Check refund progress
Account login issue	Reset password guidance

Analysis of these queries shows that nearly **75%** of customer questions are repetitive and follow predefined procedures. These repetitive tasks can be automated using Generative AI, allowing human agents to focus on more complex issues.

Existing System

Current customer support process:

Customer → Customer Care Executive → Database Search → Response → Customer

Problems

- Slow response
- Human errors
- Limited working hours
- High staffing costs
- Agent burnout
- Poor scalability

Proposed AI-Based Solution

The proposed system uses **Large Language Models (LLMs)** integrated with company databases to provide intelligent customer support.

Workflow

Customer Query

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Language Detection

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Intent Recognition

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Knowledge Base Search

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LLM Generates Response

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Customer Receives Answer

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Feedback Collection

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Continuous Improvement

Technology Stack

Component	Technology
Language Model	GPT
Intent Detection	BERT
Fine-Tuning	LoRA
Efficient Training	PEFT
Human Feedback	RLHF
Backend	Python (FastAPI/Flask)
Database	MySQL / PostgreSQL
Vector Database	FAISS / Pinecone
Frontend	React / Web Chat
Cloud Platform	AWS / Azure / GCP

Suitable Model Selection

1. GPT (Generative Pre-trained Transformer)

GPT is the primary model used to generate conversational responses.

Justification

GPT is capable of:

- Understanding natural language
- Maintaining conversation context
- Answering complex questions
- Generating human-like responses
- Supporting multiple languages

Example:

Customer:

"My order hasn't arrived."

GPT:

"I'm sorry for the delay. Let me check your order status. Please provide your Order ID."

Advantages

- Human-like conversations
- High accuracy
- Context awareness
- Easy integration

2. BERT (Bidirectional Encoder Representations from Transformers)

BERT is mainly used for:

- Intent classification
- Sentiment analysis
- Query categorization

Instead of generating responses, BERT identifies what the customer wants.

Example:

Input:

"I forgot my password."

BERT Output:

Intent = Password Reset

This helps the system route the request correctly.

Justification

BERT performs exceptionally well in:

- Text classification
- Named Entity Recognition
- Intent Detection

3. RLHF (Reinforcement Learning from Human Feedback)

RLHF improves GPT responses using human feedback.

Process:

AI Response

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Human Agent Reviews

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Feedback Collected

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Model Updated

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Better Future Responses

Justification

RLHF helps:

- Improve answer quality
- Reduce hallucinations
- Make responses more polite
- Increase customer satisfaction

4. LoRA (Low-Rank Adaptation)

Training very large language models is expensive.

LoRA allows organizations to fine-tune GPT efficiently using fewer computational resources.

Example:

Instead of retraining the entire model:

Train only a few parameters.

Benefits

- Faster training
- Lower GPU cost
- Less storage
- Easy deployment

5. PEFT (Parameter Efficient Fine Tuning)

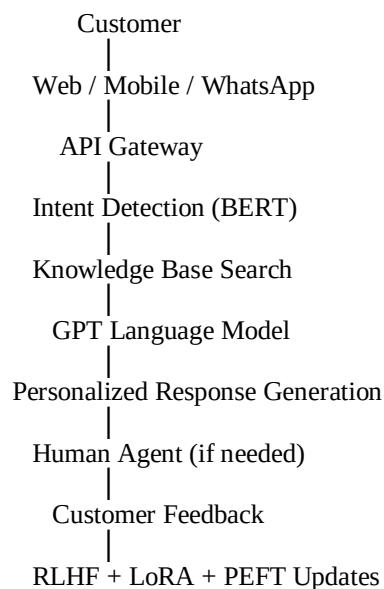
PEFT further reduces training cost.

Advantages:

- Only updates small portions of the model
- Requires less memory
- Faster fine-tuning
- Better scalability

PEFT is ideal when organizations want customized customer support without retraining the full GPT model.

System Architecture



Implementation Approach

Step 1: Data Collection

Collect historical customer support data:

- Emails
- Chat logs
- FAQs
- Call transcripts
- Product manuals
- Return policies

Step 2: Data Preprocessing

Clean the collected data.

Activities include:

- Remove duplicate records
- Remove special characters
- Tokenization
- Stop-word removal
- Language normalization

Step 3: Knowledge Base Creation

Store organizational information.

Examples:

- Shipping policy
- Return policy
- Warranty
- Product details
- Payment rules

The knowledge base is indexed using vector embeddings for efficient retrieval.

Step 4: Intent Classification

Use BERT to classify customer intent.

Examples:

Customer Query	Intent
Track order	Order Tracking
Refund money	Refund

Customer Query	Intent
Replace item	Replacement
Cancel order	Cancellation

Step 5: Response Generation

GPT receives:

- Customer query
- Intent
- Retrieved company information

GPT generates an accurate and natural response.

Step 6: Escalation

If GPT confidence is low:

Customer Query

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Transfer to Human Agent

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Agent Resolves Issue

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Conversation Stored for Future Training

Step 7: Continuous Learning

Human feedback is collected continuously.

RLHF improves response quality.

LoRA and PEFT periodically fine-tune the model using new company data.

Example Customer Interaction

Customer

"My payment failed but money was deducted."

BERT

Intent:

Payment Issue

Database

Payment status:

Pending Verification

GPT Response

"We're sorry for the inconvenience. Your payment is currently under verification. The amount is usually refunded within 3–5 business days if the transaction fails. If you need further assistance, I can connect you with a support agent."

Advantages of the Proposed Solution

- 24/7 customer support
- Faster response times
- Reduced operational costs
- Consistent responses
- Improved customer satisfaction
- Multilingual communication
- Easy scalability
- Better employee productivity
- Personalized interactions
- Reduced workload for support agents

Expected Outcomes

After implementing the AI-powered customer support automation system, the company can expect:

Metric	Before AI	After AI
Average Response Time	10–15 minutes	Less than 30 seconds

Metric	Before AI	After AI
Customer Satisfaction	75%	92–95%
Agent Workload	High	Reduced by ~60%
Operational Cost	High	Reduced by ~40%
Availability	Business Hours 24×7	
Resolution Accuracy	Moderate	High

Additional benefits include:

- Improved first-contact resolution
- Better customer retention
- Faster onboarding of support processes
- Data-driven insights into common customer issues

Limitations

Despite its advantages, the proposed solution has certain limitations:

- AI may generate incorrect or misleading responses (hallucinations).
- Complex or emotional cases still require human intervention.
- High-quality training data is essential for good performance.
- Privacy and security of customer data must be ensured.
- Integration with legacy systems can be challenging.
- Continuous monitoring and model updates are necessary.
- Initial implementation costs can be significant.
- Bias in training data may affect fairness and response quality.

Future Enhancements

The system can be further improved by adding:

- Voice-based AI customer support
- Real-time speech-to-text and text-to-speech capabilities
- Multilingual support for regional languages
- Emotion and sentiment detection for empathetic responses
- Integration with CRM and ERP systems
- Personalized recommendations based on customer history
- Predictive analytics to anticipate customer issues
- AI-powered agent assistance for faster human support

Conclusion

Generative AI and Large Language Models are revolutionizing customer support by providing intelligent, scalable, and efficient automation. In this case study, **GPT** is used for natural language generation, **BERT** for intent recognition, **RLHF** for improving response quality through human feedback, and **LoRA** with **PEFT** for efficient fine-tuning and customization. The proposed solution enables organizations to deliver faster, more consistent, and personalized support while reducing operational costs and improving customer satisfaction. Although challenges such as hallucinations, privacy concerns, and the need for human oversight remain, a well-designed hybrid approach that combines AI with human agents can significantly enhance the overall customer support experience. This makes Generative AI a powerful and practical solution for modern customer service automation.